Financial Distortions and Endogenous Technology Choice in Production Networks

Motivation

Firms are embedded in a connected production network in a closed economy. Their technology choices determine which suppliers they rely on, how exposed they are to sectoral shocks, and how shocks propagate through the economy. When production must be financed in advance, sectoral financial distortions affect firms through borrowing costs. A financially distorted sector produces less than in the efficient allocation, its price rises, and downstream firms respond by changing their technology and supplier dependence.

This project studies how financial distortions shape technology choice and aggregate network fragility. The starting point is the endogenous production-network framework of (Kopytov et al. 2024), where firms choose technologies and input shares in response to supply-chain risk. I shift the focus from uncertainty-driven technology choice to **finance-driven technology choice**: sectoral financial conditions distort production prices, and these distorted prices make private technology choices depart from the efficient network. The credit mechanism is motivated by (Wang 2025), who shows that credit-market imperfections create an additional channel of interconnection in production networks beyond the usual input-output cost channel. The quantitative and policy parts of the project are anchored in the network statistics emphasized by (Herskovic 2018), who shows that network concentration and sparsity are economically meaningful objects for aggregate risk. In this project, these statistics become policy targets: if financial shocks induce firms to rechoose technologies in ways that create excessive concentration or sparsity, sectoral taxes and subsidies can be used to lean against the resulting network fragility.

The central research question is:

> How do sectoral financial distortions shape firms' technology choices and the structure of the production network, and how should the government respond when firms' technology reallocation after financial shocks creates aggregate fragility?

The proposal has three goals. First, the static model characterizes how sectoral borrowing rates affect technology choice through comparative statics that describe how firms change their reliance on suppliers when their own or their suppliers' financial conditions change. Second, the dynamic model studies how persistent financial shocks reshape network concentration, network sparsity, and aggregate output when technology adjustment is costly. Third, the policy analysis asks whether sectoral taxes and subsidies can reduce inefficient overconcentration and make the network less vulnerable to bad shocks.

Static Environment

There are N sectors indexed by $i \in \{1, \ldots, N\}$. Sector i produces a differentiated good Y_i, which can be consumed by households or used as an intermediate input by firms in other sectors. The economy is closed.

Each sector contains a unit mass of firms $f \in [0, 1]$. Firm f in sector i produces Y_{if}. A competitive sectoral aggregator purchases firm output and produces the sectoral good according to

$$Y_i = \int_0^1 \left(\frac{Y_{if}}{\theta_i} \right)^{\theta_i} df.$$

The aggregator earns zero profit. This avoids the counterfactual implication that a bad firm-level productivity realization shuts down an entire sector. Low-productivity firms may produce zero while the sector continues to operate through other firms.

Goods market clearing requires

$$Y_i = c_i + \sum_{k=1}^{N} \int_0^1 X_{kf,i} df,$$

where c_i is household final consumption of good i, and X_{ki} is the use of good i as an intermediate input by sector k.

Firm f in sector i chooses a technology vector

$$\alpha_{if} = (\alpha_{if,1}, \ldots, \alpha_{if,N}),$$

where $\alpha_{if,j}$ governs the intensity with which firm f in sector i uses goods from sector j. Following (Kopytov et al. 2024), production combines a stochastic productivity shock, technology-specific productivity, a normalizing term useful for marginal-cost calculations, labor, and intermediate inputs:

$$Y_{if} = e^{\epsilon_{if}} A_i(\alpha_{if}) \zeta_i(\alpha_{if}) L_{if}^{\gamma_i} \prod_{j=1}^{N} X_{if,j}^{\alpha_{if,j}}.$$

The intended specification is decreasing returns to scale in measured production inputs. Therefore, the exponents satisfy

$$\gamma_i + \sum_{j=1}^{N} \alpha_{if,j} = \psi < 1.$$

The feasible technology set is therefore

$$\mathcal{A}_i = \left\{ \alpha_{if} \in \mathbb{R}_+^N : \gamma_i + \sum_{j=1}^{N} \alpha_{if,j} = \psi < 1 \right\}.$$

The firm-level productivity shock has a persistent sectoral component and an idiosyncratic firm component:

$$\epsilon_{if,t} = z_{i,t} + u_{if,t}.$$

The sectoral component $z_{i,t}$ affects sectoral output and prices. The firm-level component $u_{if,t}$ is idiosyncratic noise. With a unit mass of firms, this idiosyncratic component shapes the distribution of active firms but does not by itself generate aggregate sectoral risk.

Financial Distortion

Firms need working capital to pay labor and intermediate-input costs before sales are received. Wages are normalized to one. Firm f in sector i's borrowing need is

$$B_{if} := L_{if} + \sum_{j=1}^{N} P_j X_{if,j}.$$

All firms in sector i face the same sector-specific gross borrowing rate $R_i > R^f$, where R^f is the economy-wide safe gross rate. The spread $R_i - R^f$ is the reduced-form financial distortion in the model. It stands in for intermediation costs that creditors incur when lending to sector i: the cost of verifying a firm's realized productivity before disbursing funds, and the cost of keeping committed liquidity available for every firm in the sector even though only some of them end up drawing it. These costs are not derived from a lender's problem; the spread is taken as a sectoral primitive. A higher R_i means that working capital is more expensive in sector i, lowering production relative to the efficient allocation and raising the sector's equilibrium price.

There is no separate reduced-form borrowing-capacity parameter ϕ_i. Financial conditions enter through the price of working capital:

$$R_i \uparrow \quad \Rightarrow \quad \text{effective production cost in sector } i \uparrow .$$

Sectoral taxes or subsidies are also allowed. Let τ_i denote a reduced-form sectoral policy wedge, where $\tau_i > 0$ is a tax and $\tau_i < 0$ is a subsidy. These wedges alter firms' private returns and provide the government with instruments to correct network externalities.

Timing and Firm Problems

The period has two stages.

Stage 1: Ex Ante Borrowing Choice

Before ϵ_{if} is realized, firm f in sector i commits to a borrowing amount B_{if}. It does not choose technology at this stage. The commitment is one-sided: the firm cannot borrow more than B_{if}, but the creditor disburses the funds only after productivity is realized and only if the firm will be able to repay. Let $\mathbb{1}_{if}$ denote the indicator that firm f is funded. The firm maximizes SDF-discounted expected profit:

$$\max_{B_{if} \geq 0} \mathbb{E}\left[\Lambda \, \mathbb{1}_{if} \left((1-\tau_i) P_i Y_{if} - R_i B_{if}\right)\right],$$

where Λ is the stochastic discount factor and R_i is taken as given. Borrowing is committed before productivity is known because the firm must secure liquidity before production takes place.

This stage captures the financial distortion. A higher R_i makes working capital more expensive, reduces the optimal commitment, and affects the firm's subsequent technology choice through the liquidity available for production.

Stage 2: Ex Post Technology and Production Choice

After ϵ_{if} is realized, the creditor observes it and disburses the committed amount if and only if repayment is feasible:

$$\mathbb{1}_{if} = \mathbb{1}\left\{(1-\tau_i) P_i Y_{if}(\epsilon_{if}; B_{if}) \geq R_i B_{if}\right\}.$$

Disbursement is all-or-nothing: a funded firm draws exactly B_{if}, and an unfunded firm draws nothing and produces nothing, $Y_{if} = 0$. Because the production function is increasing in ϵ_{if}, this defines a sectoral cutoff $\bar{\epsilon}_i$ such that firms with $\epsilon_{if} \geq \bar{\epsilon}_i$ are funded and the rest are rationed. The creditor never lends to a firm that cannot repay, so no loan is ever written off; low-productivity firms exit through rationing rather than default.

A funded firm then chooses its technology vector $\alpha_{if} \in \mathcal{A}_i$, labor L_{if}, and intermediate inputs $\{X_{if,j}\}_{j=1}^{N}$, spending the full committed amount:

$$L_{if} + \sum_{j=1}^{N} P_j X_{if,j} = B_{if}.$$

It solves

$$\max_{\alpha_{if}, L_{if}, \{X_{if,j}\}} (1-\tau_i) P_i Y_{if} - R_i B_{if}$$

subject to the budget above and

$$Y_{if} = e^{\epsilon_{if}} A_i(\alpha_{if}) \zeta_i(\alpha_{if}) L_{if}^{\gamma_i} \prod_{j=1}^{N} X_{if,j}^{\alpha_{if,j}}.$$

Since $R_i B_{if}$ is sunk at this stage, the funded firm's problem reduces to maximizing output given the budget, so its technology choice depends on prices and the committed scale but not directly on R_i. The rate affects technology only through the stage-1 commitment and through equilibrium prices.

Thus financial distortions affect technology choice through the price system. A sector with a high R_i produces less and has a higher equilibrium price, making it less attractive as a supplier. Other firms then re-optimize their technology vectors, changing the production network.

Households

Households consume a Cobb-Douglas final consumption aggregator:

$$C = \prod_{i=1}^{N} c_i^{\omega_i}, \qquad \sum_{i=1}^{N} \omega_i = 1.$$

They supply labor and lend to firms. Preferences are

$$\mathbb{E}_0 \sum_{t=0}^{\infty} \beta^t \left[\log C_t - \chi \frac{L_t^{1+\varphi}}{1+\varphi} \right],$$

where N_t is aggregate labor supply, β is the discount factor, and φ governs the curvature of labor disutility.

The household budget constraint includes wage income, consumption expenditure, interest income from firm borrowing, firm profits, and rebates of taxes net of subsidy costs. With the wage normalized to one, a period budget constraint can be written schematically as

$$\sum_{i=1}^{N} P_i c_i + A' \leq L + \sum_{i=1}^{N} \int_0^1 \mathbb{1}_{if} \, R_i B_{if} \, df + \sum_{i=1}^{N} \Pi_i + T,$$

where A' denotes next-period assets, $\sum_i \Pi_i$ denotes firm profits rebated to households, and T is net fiscal rebates: tax revenues minus subsidy costs. The

sectoral gross rate R_i summarizes the financial terms at which firms in sector i obtain working capital.

The household side is intentionally kept simple. The key financial object for the firm is the sector-specific rate R_i, not a household screening rule. This follows the spirit of (Wang 2025) by letting financial conditions shape production decisions, while keeping the reduced-form financial distortion on the price side.

Network Statistics

The paper focuses on two summary statistics of the equilibrium network, following (Herskovic 2018). Let $\omega_i^Y \equiv P_i Y_i / \sum_{k=1}^N P_k Y_k$ denote sector i's share of aggregate output, and let $\alpha_{i,j} \equiv \int_0^1 \alpha_{if,j}\, df$ denote the sector-average intensity with which sector i uses good j.

Network concentration is the negative entropy of the output-share distribution:

$$\mathcal{N}^C = \sum_{i=1}^{N} \omega_i^Y \log \omega_i^Y.$$

A higher value means that output is concentrated in fewer sectors.

Sectoral sparsity is the negative entropy of sector i's input-share distribution:

$$\mathcal{N}_i^S = \sum_{j=1}^{N} \alpha_{i,j} \log \alpha_{i,j}.$$

A higher $\mathcal{N}_i^S$ means that sector i relies on fewer suppliers, so its input structure is more specialized. Network sparsity is the output-share-weighted average of sectoral sparsity:

$$\mathcal{N}^S = \sum_{i=1}^{N} \omega_i^Y \mathcal{N}_i^S = \sum_{i=1}^{N} \omega_i^Y \left(\sum_{j=1}^{N} \alpha_{i,j} \log \alpha_{i,j} \right).$$

Both statistics are endogenous in this model: financial conditions $\{R_i\}$ change technology choices $\alpha_{if,j}$, which change input shares, prices, output shares, and hence $\mathcal{N}^C$ and $\mathcal{N}^S$.

Analytical Targets

The static model is designed to study how financial distortions affect endogenous technology choice through prices and working-capital costs, and how those choices aggregate into the network statistics above. The comparative statics are organized from the aggregate level to the firm level.

Monetary Policy and Network Structure

Write the sectoral rate as $R_i = R^f + s_i$, where $s_i > 0$ is the sectoral spread. Holding spreads fixed, a change in the safe rate shifts every sector's cost of working capital at once. The first targets are

$$\frac{\partial \mathcal{N}^C}{\partial R^f} \qquad \text{and} \qquad \frac{\partial \mathcal{N}^S}{\partial R^f}.$$

These ask whether a uniform tightening of financial conditions, the monetary policy side of the model, makes the network more concentrated and more sparse. The sign is not obvious: a higher R^f raises costs in every sector, but sectors differ in γ_i, in their position in the network, and in their sensitivity to working-capital costs, so output shares and input shares need not move proportionally.

Sectoral Financial Shocks and Input Structure

The second target is the response of a sector's own input structure to its own financial condition:

$$\frac{\partial \mathcal{N}_i^S}{\partial R_i}.$$

This captures how a sector reorganizes its input mix when it faces a sectoral financial shock. A higher R_i raises the cost of every unit of working capital in sector i, which can push firms toward technologies that economize on purchased inputs or toward a narrower set of suppliers.

Propagation Through Technology Choice

The remaining targets are at the firm level and concern the input intensity $\alpha_{if,j}$ of firm f in sector i on good j. Sector-average responses follow by integrating over f.

Downstream propagation:

$$\frac{\partial \alpha_{if,j}}{\partial R_j}.$$

This captures how supplier j's financial condition affects the technology choice of its customers. If R_j rises, sector j's production cost and price rise, making good j less attractive as an input, so the financial shock in j travels downstream through the input choices of firms in i.

Upstream propagation:

$$\frac{\partial \alpha_{if,j}}{\partial R_i}.$$

This captures how a buyer's own financial condition changes its reliance on supplier j. A higher R_i raises the cost of working capital for firms in sector i, lowers their optimal production scale, and changes which technologies are feasible or profitable after shocks are realized. The resulting change in demand for good j transmits the financial shock in i upstream to its suppliers.

Interaction between own and upstream financial conditions:

$$\frac{\partial^2 \alpha_{if,j}}{\partial R_i \partial R_j}.$$

This cross derivative asks whether a sector's own financing cost changes how sensitive its technology choice is to supplier j's financing cost. It therefore measures whether a financial shock in sector i amplifies or dampens the propagation of financial shocks arriving from upstream.

Dynamic Extension

The next step is to introduce persistent sectoral financial conditions. The productivity process is decomposed into a persistent sectoral component and iid firm-level noise:

$$z_{i,t} = \rho_i^z z_{i,t-1} + \eta_{i,t}^z,$$

and

$$u_{if,t} \sim F_i,$$

where $u_{if,t}$ is iid across firms and time. The firm-level component generates cross-sectional dispersion in production outcomes, while the persistent sectoral component generates aggregate sectoral productivity movements.

Financial conditions are summarized by a sectoral intermediation wedge $s_{i,t}$, the reduced-form counterpart of verification and committed-liquidity costs, which follows

$$s_{i,t} = \rho_i^s s_{i,t-1} + \eta_{i,t}^s.$$

The sectoral gross borrowing rate is

$$R_{i,t} = R_t^f + s_{i,t},$$

where R_t^f is the economy-wide safe gross rate. An adverse financial shock is $\eta_{i,t}^s > 0$, which raises $s_{i,t}$ and therefore raises $R_{i,t}$. Importantly, $R_{i,t}$ does not

load directly on the productivity shock $z_{i,t}$. Productivity shocks and financial shocks are separate primitives in the decomposition.

Technology adjustment is costly. Define

$$d_{if,t} = \|\alpha_{if,t} - \alpha_{if,t-1}\|_2 ,$$

and let the adjustment cost be

$$\kappa d_{if,t}^2.$$

The dynamic firm problem then trades off the benefit of adapting input shares to current financial and productivity conditions against the cost of changing production technology. This friction creates persistent network responses to persistent financial shocks.

Quantitative Exercise

The quantitative model will study how productivity shocks and financial shocks affect aggregate output and the structure of the production network. The network statistics defined above are tracked period by period:

$$\mathcal{N}_t^C = \sum_{i=1}^{N} \omega_{i,t}^Y \log \omega_{i,t}^Y, \qquad \mathcal{N}_{i,t}^S = \sum_{j=1}^{N} \alpha_{i,j,t} \log \alpha_{i,j,t}, \qquad \mathcal{N}_t^S = \sum_{i=1}^{N} \omega_{i,t}^Y \mathcal{N}_{i,t}^S.$$

The main impulse-response exercises are:

1. An adverse productivity shock to a sector, modeled as a negative shock to $\eta_{i,t}^z$, which lowers the persistent sectoral productivity component $z_{i,t}$.
2. An adverse financial shock to a sector, modeled as a positive shock to $\eta_{i,t}^s$, which raises the sectoral intermediation wedge $s_{i,t}$ and therefore raises the gross borrowing rate $R_{i,t}$.
3. A monetary tightening, modeled as a rise in the safe rate R_t^f with sectoral spreads $s_{i,t}$ held fixed, which raises $R_{i,t}$ in every sector at once.

For each shock, the quantitative exercise will trace impulse responses of:

1. Aggregate output.
2. Network concentration $\mathcal{N}_t^C$.
3. Sectoral sparsity $\mathcal{N}_{i,t}^S$ and network sparsity $\mathcal{N}_t^S$.
4. Sectoral technology choices $\alpha_{i,j,t}$.
5. Sectoral prices, borrowing, and financial spreads.

Externality and Industrial Policy

The model creates a natural externality. Individual firms choose technologies based on private expected profits, distorted sectoral prices, and sector-specific borrowing costs. They internalize how input choices affect their own production costs, but they do not internalize how their technology choices affect the fragility of the aggregate production network.

In particular, private technology choices may generate excessive concentration around central suppliers. Such concentration can be privately efficient in normal times because central suppliers may be productive or cheap. But it can make the whole network vulnerable if the central sector is hit by an adverse productivity shock or a worsening of financial conditions. The planner therefore faces a tradeoff between static production efficiency and dynamic network resilience.

This opens the door to optimal industrial policy. Sectoral taxes and subsidies can be used to discourage excessive dependence on fragile central sectors or to encourage diversification toward suppliers that improve aggregate resilience. The policy problem is therefore not simply to subsidize productive sectors, but to account for how sectoral interventions reshape the endogenous network and the economy's exposure to bad shocks.

The central policy question is:

> How should the government choose sectoral taxes and subsidies to reduce inefficient network overconcentration without sacrificing too much aggregate output?

The answer depends on the interaction between financial distortions, technology choice, and network centrality. This interaction is precisely what the model is designed to make transparent.

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